Assignment 2: Sentiment Analysis on Electric Vehicles (EVs)

---

1. Problem Statement

Electric Vehicles (EVs) play a crucial role in India's transition toward sustainable and eco-friendly transportation. Public opinion regarding EVs on social media platforms such as X (Twitter) includes a mix of positive sentiments (environmental benefits and innovation), neutral opinions, and negative concerns (charging infrastructure, cost, and range anxiety).

The problem is to perform sentiment analysis on 100 public tweets and classify them into **positive, negative, or neutral categories** to understand the overall public outlook.

---

2. Objective

- To collect 100 tweets related to Electric Vehicles.
- To manually label each tweet as **positive, negative, or neutral**.
- To apply machine learning models such as:
  - Naïve Bayes
  - Support Vector Machine (SVM)
  - Logistic Regression
- To evaluate and compare the performance of these models.
- To identify the most effective classifier for sentiment analysis.

---

3. Dataset Description

- **Source:** Tweets collected using keywords like *"EV India," "Charging Stations," and "Electric Mobility"*.
- **Features:**
  - Tweet (raw text)
  - Sentiment (manually labeled)
- **Dataset Size:** 100 tweets

---

4. Methodology

4.1 Data Preprocessing

- Removed special characters such as URLs, hashtags, and mentions (@).
- Converted all text to lowercase for uniformity.
- Removed stopwords to retain meaningful words contributing to sentiment.

4.2 Data Splitting

- Training Data: 80% (80 tweets)
- Testing Data: 20% (20 tweets)

4.3 Feature Extraction

- Used **TF-IDF Vectorization** to convert textual data into numerical form.

4.4 Model Building

The following machine learning models were trained:

- Multinomial Naïve Bayes
- Support Vector Machine (SVM)
- Logistic Regression

4.5 Evaluation Metrics

- Precision
- Recall

---

5. Results

Model Performance Comparison

Classifier	Precision	Recall
Naïve Bayes	1.0	1.0
SVM	1.0	1.0
Logistic Regression	1.0	1.0

```
... Loading Dataset...
    Train size: 80 | Test size: 20

    Evaluation (Weighted Precision & Recall) ---

    ========== Naïve Bayes ==========
                  precision    recall  f1-score   support

        Negative       1.00      1.00      1.00         6
         Neutral       1.00      1.00      1.00         8
        Positive       1.00      1.00      1.00         6

        accuracy                           1.00        20
       macro avg       1.00      1.00      1.00        20
    weighted avg       1.00      1.00      1.00        20


    ========== SVM ==========
                  precision    recall  f1-score   support

        Negative       1.00      1.00      1.00         6
         Neutral       1.00      1.00      1.00         8
        Positive       1.00      1.00      1.00         6

        accuracy                           1.00        20
       macro avg       1.00      1.00      1.00        20
    weighted avg       1.00      1.00      1.00        20


    ========== Logistic Regression ==========
                  precision    recall  f1-score   support

        Negative       1.00      1.00      1.00         6
         Neutral       1.00      1.00      1.00         8
        Positive       1.00      1.00      1.00         6

        accuracy                           1.00        20
       macro avg       1.00      1.00      1.00        20
    weighted avg       1.00      1.00      1.00        20


    🏆 Final Model Comparison (Precision & Recall):
            Classifier  Precision  Recall
           Naïve Bayes        1.0     1.0
                   SVM        1.0     1.0
    Logistic Regression       1.0     1.0
```

Observations

- All three models achieved **perfect precision and recall (1.0)**.
- The dataset had clear and distinguishable sentiment patterns.
- Models were able to classify sentiments without error on the test data.

---

6. Insights

- The high accuracy suggests that the dataset was **well-structured and clearly labeled**.
- Positive tweets mainly reflected **environmental benefits and innovation**.
- Negative tweets focused on **charging infrastructure and cost concerns**.
- Neutral tweets provided **informational or mixed opinions**.

---

7. How to Run the Program

Install required libraries

```
pip install pandas scikit-learn nltk


# Open and run the Jupyter Notebook

RollNo_Name_Assignment2.ipynb
```

---

8. Conclusion

All three machine learning models—Naïve Bayes, SVM, and Logistic Regression—performed equally well, achieving 100% accuracy on the test dataset. While all models are effective, **SVM is generally preferred** for handling high-dimensional text data.

The analysis indicates that public sentiment towards Electric Vehicles in India is **largely distinguishable**, highlighting both enthusiasm for innovation and concerns regarding infrastructure.

---

9. Student Details

- **Name:** Mohammad Zohair Shaikh
- **UIN:** 231A043
- **Roll No:** 62
- **Class:** TE (2025–26)
- **Course:** Data Analytics and Visualisation